SMARTBRIDGE - Solution Architecture

- Date: 21/09/2026
- Team ID: SWTID-2026-4223
- Team Size: 4
- Team Leader: Sweatha P
- Team Members:Lakshmi Narayanan R, Rithika P, Sandhiya K
- Project Name: AI FitTrack

| Maximum Marks: 5 Marks

Solution Architecture Diagram Structural Layout

Presentation / Client Layer → API Gateway / Express Server → Auth Service + Core Logic Service → External API Integration (Google Gemini AI API) + Data / Storage Layer (MongoDB Atlas).

Presentation Layer

Serves as the entry point for users to log in, input workout metrics, search history, and receive AI workout recommendations. Technologies: Postman, Thunder Client, React.js / Mobile Client.

API Gateway

Central HTTP server listener that routes requests, handles CORS policy, parses JSON, and forwards calls to controllers. Technology: Express.js Gateway (server.js, app.js).

Auth Service

Verifies user credentials, hashes passwords, generates signed JWTs, and authorizes protected routes via middleware. Technologies: jsonwebtoken, bcryptjs, authMiddleware.js.

Core Logic Service

Executes workout CRUD logic, performs search queries, calculates stats, and formats JSON API output. Technologies: Express Controllers (workoutController.js, aiController.js).

External API Integration

Communicates with Google Gemini Generative AI to generate tailored weekly workout plans and progress insights. Technology: Google Gemini API (@google/genai, gemini-2.5-flash).

Database Layer

Stores persistent user credentials, profile records, and complete workout logs in structured NoSQL collections. Technologies: MongoDB Atlas Cloud, Mongoose ODM (User.js, Workout.js).